

AU Airbnb Data Pipeline

Big Data | 3rd Assignment

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STACK:

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UTS

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Introduction

Australia's dynamic accommodation market, significantly influenced by platforms like Airbnb, presents a rich environment for data analysis. A comprehensive understanding of this market requires combining granular rental listing data with broader socio-economic factors. To achieve this, two key datasets are utilized: detailed Airbnb listings for Sydney and comprehensive demographic information from the Australian Census. The processing and modeling of this information are handled by a modern data stack, employing Apache Airflow for workflow orchestration and dbt Cloud for data transformation.

The primary aim is to construct a production-ready ELT (Extract, Load, Transform) data pipeline. Using a Medallion Architecture, the pipeline will systematically ingest and refine raw data through Bronze, Silver, and Gold layers within a data warehouse. The ultimate goal is to produce a unified, clean, and analysis-ready data mart, enabling ad-hoc analytical queries to address key business questions about Sydney's rental landscape.

Datasets

1. Airbnb Listings Data

This is a collection of monthly snapshots of Airbnb listings in Sydney, covering the period from **May 2020 to April 2021**. Each file (e.g., `05_2020.csv`) represents one month of data. This dataset provides a granular view of the short-term rental market, capturing property attributes, pricing, availability, and host information. It is the primary source for analyzing rental trends and market dynamics.

Table 1 Airbnb Listings Data Dictionary

Column Name	Description
id	Unique identifier for the listing.
listing_url	URL to the Airbnb listing page.
name	The title of the listing.
description	Detailed description of the property.
host_id	Unique identifier for the host.
host_name	Name of the host.
suburb	The suburb where the property is located.
property_type	Type of property (e.g., 'Apartment', 'House').
room_type	Type of room (e.g., 'Entire home/apt', 'Private room').
accommodates	Maximum number of guests the listing can accommodate.
bathrooms_text	Description of bathroom facilities.
bedrooms	Number of bedrooms.

Column Name	Description
beds	Number of beds.
amenities	A list of amenities offered.
price	The nightly price in AUD.
minimum_nights	The minimum stay required.
maximum_nights	The maximum stay allowed.
number_of_reviews	Total number of reviews for the listing.
review_scores_rating	The overall rating from reviews (out of 5 or 100).

2. Census Demographics Data

Sourced from the **2016 Australian Census**, this data provides a socio-economic profile of various Local Government Areas (LGAs) in New South Wales. It is used to enrich the Airbnb data, allowing for deeper analysis by correlating rental trends with community demographics. The data is divided into two key files:

- **G01 (Selected Person Characteristics by Sex):** Contains population counts broken down by characteristics like age, marital status, and education for each LGA.

Table 2 Selected Person Characteristics by Sex Data Dictionary

Column Name	Description
LGA_CODE_2016	The official code for the Local Government Area.
Tot_P_M	Total number of males.
Tot_P_F	Total number of females.
Tot_P_P	Total number of persons.
Age_0_4_yr_M	Number of males aged 0-4 years.
...	*Columns continue for various age, sex, and other demographic categories.*

- **G02 (Selected Medians and Averages):** Provides median and average values for key demographic indicators like age, income, and mortgage or rent payments per LGA.

Table 3 Selected Medians and Averages Data Dictionary

Column Name	Description
LGA_CODE_2016	The official code for the Local Government Area.
Median_age_persons	Median age of all persons.

Column Name	Description
Median_mortgage_repay_monthly	Median monthly mortgage repayment.
Median_rent_weekly	Median weekly rent payment.
Median_tot_prsnl_inc_weekly	Median total personal income per week.
Median_tot_fam_inc_weekly	Median total family income per week.
Average_hhld_size	Average household size.

3. Geospatial Lookup Tables

These files serve as supplementary data to map suburbs to their corresponding Local Government Areas (LGAs), which is essential for joining the Airbnb and Census datasets.

- **NSW_LGA_CODE.csv:** A reference table mapping LGA names to their official codes.
- **NSW_LGA_SUBURB.csv:** A mapping file that links NSW suburbs to their respective LGA.

Table 4 NSW_LGA_CODE & NSW_LGA_SUBURB Data Dictionary

File Name	Column Name	Description
NSW_LGA_CODE.csv	LGA_CODE	The official code for the LGA.
	LGA_NAME	The full name of the LGA (e.g., 'City of Sydney').
NSW_LGA_SUBURB.csv	LGA_NAME	The full name of the LGA.
	SUBURB_NAME	The name of the suburb.

High level Architecture

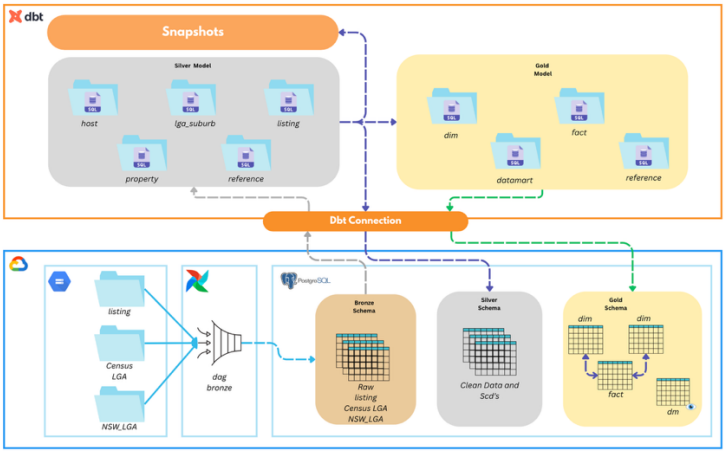


Figure 1 High Level Architecture

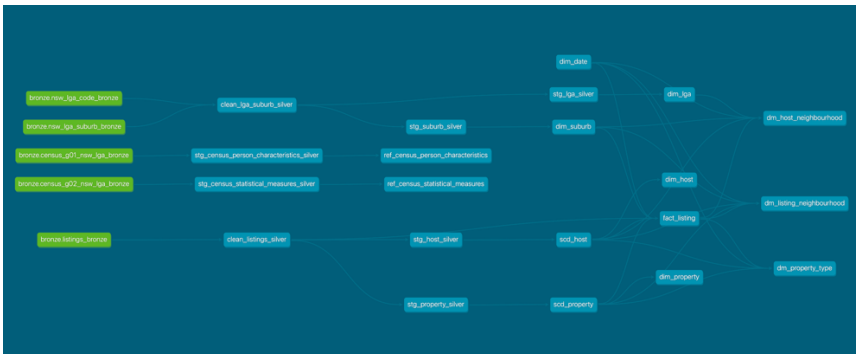


Figure 2 Dbt Pipeline

Bronze Layer

At the outset of the project, three distinct database schemas—bronze, silver, and gold—were established within the **PostgreSQL** database instance running on **Google Cloud Platform (GCP)**. This separation implements the Medallion Architecture, providing clear boundaries for data as it moves from raw to refined.

The Bronze Layer is the foundational stage of this architecture, acting as the raw data vault. All the source files for the project are first stored in a **Google Cloud Storage (GCS)** bucket, which serves as the single source of truth.

The process of loading this data into the PostgreSQL database is orchestrated by an **Apache Airflow** instance, also hosted on GCP. The dag_bronze.py DAG defines the tasks for this "load" step. This DAG is responsible for taking the source files from the GCS bucket and loading them into their respective tables within the bronze schema.

A key decision was made during this stage regarding the Airbnb data. While all Census files (2016Census_G01_NSW_LGA.csv, 2016Census_G02_NSW_LGA.csv) and lookup tables (NSW_LGA_CODE.csv, NSW_LGA_SUBURB.csv) were loaded, **only the first month of Airbnb listings (05_2020.csv) was loaded into the Bronze Layer.**

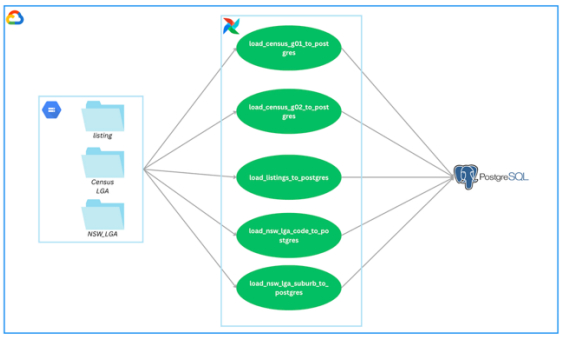


Figure 3 Bronze Airflow DAG

Silver Layer

This layer handles all data cleansing, type casting, renaming, and joins necessary to make the data reliable for analysis.

Materialization and Transformation Strategy

The Silver Layer uses several different materialization strategies, as defined by the dbt models:

- **Incremental Models (incremental):** This is a key optimization. Key models, including stg_host_silver, stg_property_silver, and the final clean_listings_silver, are materialized as incremental. This means on subsequent runs, dbt only processes new or changed records.
- **Full-Refresh Models (table):** Models like clean_lga_suburb_silver are fully dropped and recreated on every dbt run.
- **Snapshot Models (snapshot):** The scd_host and scd_property models are used to implement Slowly Changing Dimensions (SCD), which track the history of changes to records over time.

Cleaning the Sources

These models run first. They interface directly with the bronze schema, performing basic cleaning and renaming.

- **Model:** clean_listings_silver.sql
  - **Source:** Listing Data
  - **Purpose:** It takes the *incrementally staged* property data and performs in-depth transformations, including:
    - **Price Cleansing:** Converting the price column (text) into a numeric (float or decimal) data type.
    - **Null Handling:** Coalescing null values in critical fields like bedrooms or bathrooms to a default value (e.g., 1).
    - **Data Type Casting:** Ensuring all dates, booleans, and numeric fields are in the correct format for analysis.
  - **Table Created:** silver.clean_listings_silver
    - This is the final, clean, incremental table of property listings, ready for host and property models.

- **Model:** stg_host_silver.sql
  - **Source:** silver.clean_listings_silver
  - **Purpose:** To extract and de-duplicate host-specific information from the raw listings data. It selects columns like host_id, host_name, and host_since, renames them, and casts them to the correct data types. This model is

materialized as **incremental**, meaning dbt will only process new or updated host records on subsequent runs.
    - **Table Created:** silver.stg_host_silver
        - This table serves as a clean, incremental source for host.
- **Model:** stg_property_silver.sql
    - **Source:** silver.clean_listings_silver
    - **Purpose:** To extract all property-specific columns (like id as listing_id, suburb, property_type, price, bedrooms, etc.) from the raw listings. This model is responsible for the initial column renaming and basic type casting. Like the host model, this is also **incremental**, allowing the pipeline to efficiently stage only new or changed properties.
    - **Table Created:** silver.stg_property_silver
        - This table acts as the initial "staged" version of the property data, updated incrementally.
- **Model:** clean_lga_suburb_silver.sql
    - **Source(s):** bronze.nsw_lga_code_bronze and bronze. nsw_lga_suburb_bronze
    - **Purpose:** To join the staged LGA and suburb tables, creating a single, reliable lookup table that maps each suburb to its official lga_code and lga_name. This is materialized as a table (full-refresh).
    - **Table Created:** silver.clean_lga_suburb_silver
- **Model:** stg_lga_silver.sql
    - **Source:** silver.clean_lga_suburb_silver
    - **Purpose:** To clean the LGA code lookup table. It renames columns (e.g., LGA_CODE to lga_code, LGA_NAME to lga_name) and casts types.
    - **Table Created:** silver.stg_lga_silver
        - A clean lookup table for LGA codes.
- **Model:** stg_suburb_silver.sql
    - **Source:** silver.clean_lga_suburb_silver
    - **Purpose:** To clean the suburb-to-LGA mapping table, renaming columns for consistency.
    - **Table Created:** silver.stg_suburb_silver
        - A clean lookup table for suburb-to-LGA relationships.
- **Model:** stg_census_person_characteristics_silver.sql
    - **Source:** bronze. census_g01_nsw_lga_bronze
    - **Purpose:** To stage the Census G01 (Person Characteristics) data. This is step involves renaming the original columns to lower case and LGA_CODE_2016 to just lga_code.
    - **Table Created:** silver.stg_census_person_characteristics_silver
- **Model:** stg_census_statistical_measures_silver.sql
    - **Source:** bronze. census_g02_nsw_lga_bronze
    - **Purpose:** To stage the Census G02 (Medians and Averages) data. This is step involves renaming the original columns to lower case and LGA_CODE_2016 to just lga_code.
    - **Table Created:** silver.stg_census_statistical_measures_silver

Slowly Changing Dimension (SCD) Implementation

Dbt snapshots are used to build historical tables that track changes to key entities. These models create the final two tables in the schema.

- **Model:** scd_host.sql
  - **Source:** silver.stg_host_silver
  - **Purpose:** To implement a Type 2 SCD. This snapshot monitors the stg_host_silver table. Based on the host_id, it tracks changes in the other columns, and using scraped_date as the reference date for the effective dates.
  - **Table Created:** silver.dim_host_scd
    - This historical table contains dbt_valid_from and dbt_valid_to columns, allowing an analyst to see what a host's details were at any point in time.
- **Model:** scd_property.sql
  - **Source:** silver.stg_property_silver
  - **Purpose:** To implement a Type 2 SCD on the final clean listings. This snapshot monitors the stg_host_silver table. Based on the host_id, it tracks changes in the other columns, and using scraped_date as the reference date for the effective dates.
  - **Table Created:** silver.dim_property_scd
    - This table captures the complete history of every listing, enabling point-in-time analysis of the properties.

Gold Layer

This layer creates aggregated, analysis-ready tables and views. It uses the clean data from the Silver Layer to build a **star schema**, which is then used to create several denormalized **data marts** tailored for specific business questions.

Dimension and Reference Tables

These models create the descriptive tables that provide context for the business metrics.

- **Model:** dim_date.sql
  - **Source:** N/A (Generated)
  - **Purpose:** To create a comprehensive date dimension table. This is a utility model that programmatically generates a record for each day within a specified date range. It enriches each date with useful attributes like day_of_week, month_name, quarter, year, and flags for is_weekend. This table is essential for performing any time-based analysis, filtering, or aggregation.
  - **Table Created:** gold.dim_date

- **Model:** dim_host.sql
  - **Source:** silver.dim_host_scd
  - **Purpose:** To create a clean, current-state dimension table of hosts by selecting only the records that are currently active (where dbt_valid_to IS NULL) from the historical SCD table.
  - **Table Created:** gold.dim_host
- **Model:** dim_property.sql
  - **Source:** silver.dim_property_scd
  - **Purpose:** To create a current-state dimension of properties by selecting the latest version of each property from its historical SCD table.
  - **Table Created:** gold.dim_property
- **Model:** dim_lga.sql
  - **Source:** silver.clean_lga_suburb_silver
  - **Purpose:** To create a distinct list of Local Government Areas (LGAs) from the silver lookup table.
  - **Table Created:** gold.dim_lga
- **Model:** dim_suburb.sql
  - **Source:** silver.clean_lga_suburb_silver
  - **Purpose:** To create a distinct list of suburbs, including the lga_code as a foreign key to the dim_lga table.
  - **Table Created:** gold.dim_suburb
- **Model:** ref_census_person_characteristics.sql
  - **Source:** silver.stg_census_person_characteristics_silver
  - **Purpose:** To promote the cleaned Census G01 (population) data into the Gold layer, making it available for joins in the final data marts.
  - **Table Created:** gold.ref_census_person_characteristics
- **Model:** ref_census_statistical_measures.sql
  - **Source:** silver.stg_census_statistical_measures_silver
  - **Purpose:** To create a clean reference table for the Census G02 (statistical measures) data, making metrics like median income and rent directly accessible.
  - **Table Created:** gold.ref_census_statistical_measures

Fact Table

This model creates the central fact table, which contains the quantitative metrics and links all the dimension tables together.

- **Model:** fact_listing.sql
  - **Source(s):** silver.clean_listings_silver and all the gold.dim_* tables.
  - **Purpose:** To build the transactional core of the data model. This table contains foreign keys (e.g., a date_key for dim_date, host_id, property_id, lga_code) to each dimension and holds the measurable facts about each listing, such as price, number_of_reviews, and accommodates.
  - **Table Created:** gold.fact_listing

Data Mart

They join the fact and dimension tables and perform aggregations to create wide, denormalized views.

- **Model:** dm_listing_neighbourhood.sql
  - **Source(s):** gold.fact_listing, gold.dim_property, gold.dim_lga, gold.ref_census_* tables.
  - **Purpose:** To create an aggregated view of the listing market at the LGA (neighbourhood) level. It calculates key metrics like number_of_active_listings and average_price per LGA, enriching this with census data such as median_income_of_lga.
  - **View Created:** gold.dm_listing_neighbourhood
- **Model:** dm_host_neighbourhood.sql
  - **Source(s):** gold.fact_listing, gold.dim_host, gold.dim_lga.
  - **Purpose:** To create an aggregated view of host activity by neighbourhood.
  - **View Created:** gold.dm_host_neighbourhood
- **Model:** dm_property_type.sql
  - **Source(s):** gold.fact_listing, gold.dim_property.
  - **Purpose:** To analyze the market by property category. This model groups listings by property_type (e.g., 'Apartment', 'House') and calculates KPIs for each.
  - **View Created:** gold.dm_property_type

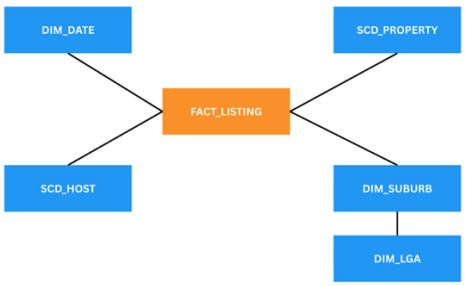


Figure 4 Star Schema Using SCD Tables

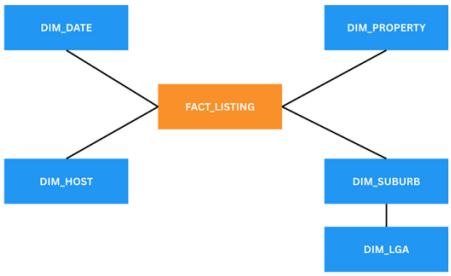


Figure 5 Star Schema Using DIM Tables

The Gold Layer's star schema was intentionally designed for analytical flexibility. The central fact_listing table can be joined with two types of tables:

1. **Current State Analysis:** The fact table can point to the dim_* tables, so it can get the latest state of the data.
2. **Historical Analysis:** The fact table can also point to the scd tables. This allows to reconstruct what the data looked like at any specific date in the past.

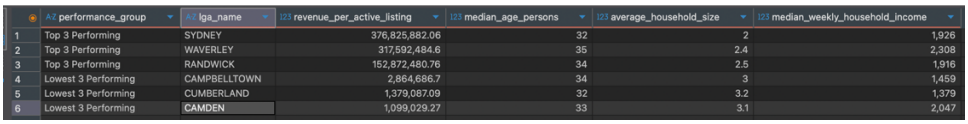
*After the initial setup and successful processing of the first month of data (05_2020.csv), the ELT pipeline was executed **manually for each subsequent month**. This iterative*

process involved running the full pipeline for the June 2020 data, then July 2020, and so on, until the final dataset for April 2021 was processed.

Findings

This section presents the key business and analytical questions addressed by the project, along with the insights derived from the data analysis.

1. **What are the demographic differences (e.g., age group distribution, household size) between the top 3 performing and lowest 3 performing LGAs based on estimated revenue per active listing over the last 12 months?**

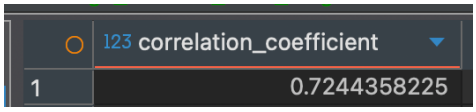


	performance_group	lga_name	revenue_per_active_listing	median_age_persons	average_household_size	median_weekly_household_income
1	Top 3 Performing	SYDNEY	376,825,882.06	32	2	1,926
2	Top 3 Performing	WAVERLEY	317,592,484.6	35	2.4	2,308
3	Top 3 Performing	RANDWICK	152,872,480.76	34	2.5	1,916
4	Lowest 3 Performing	CAMPBELLTOWN	2,864,686.7	34	3	1,459
5	Lowest 3 Performing	CUMBERLAND	1,379,087.09	32	3.2	1,379
6	Lowest 3 Performing	CAMDEN	1,099,029.27	33	3.1	2,047

Figure 6 Question 1 Results

   - **Location is the Ultimate Driver of Revenue:** The most critical factor for Airbnb success is proximity to tourist and business hubs like the Central Business District and iconic beaches.
   - **Demographics Define the Market Type:** The top-performing areas are characterized by smaller household sizes and younger populations, typical of transient, professional, and tourist-heavy zones.

2. **Is there a correlation between the median age of a neighbourhood (from Census data) and the revenue generated per active listing in that neighbourhood?**



	correlation_coefficient
1	0.7244358225

Figure 7 Question 2 Results

   - **Strong Positive Correlation:** A correlation coefficient of 0.72 indicates a **strong positive relationship** between the median age of a neighbourhood and the revenue per active listing. This means that, generally, as the median age of an LGA increases, the revenue generated by Airbnb listings in that area also tends to increase.

3. **What will be the best type of listing (property type, room type and accommodates for) for the top 5 "listing_neighbourhood" (in terms of estimated revenue per active listing) to have the highest number of stays?**

	listing_neighbourhood	property_type	room_type	accommodates	estimated_revenue_per_active_listing	total_number_of_stays
1	SYDNEY	ENTIRE APARTMENT	ENTIRE HOME/APT	2	376,825,882.06	390,723
2	WAVERLEY	ENTIRE APARTMENT	ENTIRE HOME/APT	2	317,592,484.6	209,664
3	RANDWICK	ENTIRE APARTMENT	ENTIRE HOME/APT	4	152,872,480.76	109,966
4	WOOLLAHRA	ENTIRE APARTMENT	ENTIRE HOME/APT	2	115,302,006.88	57,128
5	NORTH SYDNEY	ENTIRE APARTMENT	ENTIRE HOME/APT	2	61,727,549.19	65,938

Figure 8 Question 3 Results

- **The "Entire Apartment" is top**: In all five of the highest-revenue LGAs, the single most successful property type is overwhelmingly an **"Entire apartment"** offered as an **"Entire home/apt"**.
- **The Ideal Guest is a Couple or Solo Traveler:** For four out of the five top neighbourhoods—**Sydney, Waverley, Woollahra, and North Sydney**—the optimal accommodation capacity is **two guests**.
- **The Randwick Family & Group Exception:** is the notable outlier, with the highest number of stays coming from listings that accommodate **four guests**.

4. **For hosts with multiple listings are their properties concentrated within the same LGA, or are they distributed across different LGAs?**

	host_strategy	number_of_hosts	pct_of_multi_listing_hosts	pct_of_all_hosts
1	Concentrated in one LGA	3,067	97.12	9.9
2	Distributed across multiple LGAs	91	2.88	0.29

Figure 9 Question 4 Results

- **Hyper-Local Concentration is the Norm:** An astounding **97.12%** of hosts with multiple listings choose to keep all their properties within a **single LGA**. This indicates that the typical multi-listing host is not a large, geographically diverse corporation, but rather an individual or small business focused on a specific neighbourhood.
- **Distributed Hosts are a Niche Minority:** The hosts who distribute their properties across multiple LGAs represent a tiny fraction of the market (**2.88%** of multi-listers and just **0.29%** of all hosts).

5. **For hosts with a single Airbnb listing does the estimated revenue over the last 12 months cover the annualised median mortgage repayment in the corresponding LGA? Which LGA has the highest percentage of hosts that can cover it?**

	lga_name	hosts_covering_mortgage	total_single_listing_hosts	coverage_percentage
1	ARMIDALE REGIONAL	2	2	100
2	UPPER HUNTER SHIRE	73	101	72.28
3	NORTHERN BEACHES	812	1,204	67.44
4	BATHURST REGIONAL	4	6	66.67
5	DUNGOG	26	43	60.47
6	MOSMAN	129	228	56.58
7	SUTHERLAND SHIRE	61	116	52.59
8	WAVERLEY	1,417	2,739	51.73
9	SYDNEY	1,774	3,575	49.62
10	RANDWICK	770	1,582	48.67

Figure 10 Question 5 Results

- **Not enough coverage:** While **Armidale Regional** demonstrates that it's possible to achieve **100% mortgage coverage**, it does so with a very small number of hosts, making it an outlier. More significantly, even the other LGAs in the top 10—including high-revenue areas like the **Northern Beaches (67%)** and **Waverley (52%)**—fall well short of the 100% mark. This clearly indicates that in the vast majority of cases, even for successful hosts in premium locations, Airbnb income alone is often insufficient to cover the entire median mortgage payment.

References

- Australian Bureau of Statistics. (2021). *Census DataPacks*. Accessed on October 19, 2025, from https://www.abs.gov.au/census/find-census-data/datapacks
- Inside Airbnb. (n.d.). *Get the Data*. Accessed on October 19, 2025, from http://insideairbnb.com/get-the-data/